Global Deforestation and Forest Cover Trend Analysis Using Python

1. Introduction

Forests play a vital role in maintaining ecological balance, supporting biodiversity, and regulating climate change. However, increasing industrialization, urbanization, and agricultural expansion have contributed significantly to global deforestation. This project aims to analyze global forest area trends using World Bank datasets and explore whether GDP growth has any relationship with forest area changes.

2. Objective

The main objectives of this project are:

- To analyze global forest area trends from 1992 to 2021.
- To identify countries with the highest forest area.
- To determine countries experiencing major forest loss and forest gain.
- To study the relationship between GDP growth and forest area using correlation analysis.
- To generate insights through Exploratory Data Analysis (EDA) and data visualization.

3. Tools and Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Jupyter Notebook

4. Dataset Description

The datasets used in this project were obtained from World Bank data sources.

Datasets Used:

- Forest Area Dataset (sq. km)
- GDP Growth Dataset

The datasets contain country-wise yearly data covering multiple years and were cleaned and transformed into long format for analysis.

5. Methodology

The following steps were performed during the analysis:

1. Loaded and inspected datasets using Pandas.
2. Cleaned missing values and removed aggregate regional entries.
3. Converted wide-format datasets into long-format structure using the melt() function.
4. Performed trend analysis and country-level comparisons.
5. Created visualizations to study forest trends and GDP relationships.
6. Conducted correlation analysis between GDP growth and forest area.

6. Key Findings

- Global forest area showed a steady decline between 1992 and 2021, indicating long-term deforestation trends.

- Russian Federation, Brazil, Canada, and the United States had the largest forest areas globally.
- Brazil experienced the highest forest loss since 1990, highlighting severe deforestation in the Amazon region.
- Some countries demonstrated forest gain, suggesting successful conservation and reforestation initiatives.
- Correlation analysis between GDP growth and forest area produced a very weak negative correlation (-0.027), indicating that economic growth alone does not strongly explain forest loss.

7. Conclusion

The analysis highlights that global deforestation remains a significant environmental challenge. While economic growth may indirectly contribute to forest depletion, the weak correlation result suggests that forest area changes are influenced by multiple factors such as environmental policies, land-use management, industrialization, and conservation efforts. This project demonstrates how data analytics and visualization can help understand environmental trends and support sustainable decision-making.

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Tools Used:
Python | Pandas | Matplotlib | Seaborn | Jupyter Notebook